Requirement Specification And Design Document

Course Registration System

Table of Contents

I. Functional Requirements 2

II. Non-Functional Requirements 4

III. Data Flow Diagram 6

IV. Use case Diagram 7

V. Class Diagram 26

VI. Data Model 27

VII. Interface Design Description 28

I. Functional Requirements

1. User Authentication

- FR01: The system shall allow students, faculty members, and administrators to log in using their username (Student ID / Lecturer or Administrator ID) and password.
- FR02: The system shall identify the user's role (Student, Lecturer, Admin) after login and redirect them to the corresponding dashboard.
- FR03: The system shall refuse access and display an error message if the username or password is incorrect.
- FR04: The system shall block login attempts for users whose accounts are locked or suspended.
- FR05: The system shall allow users to change their password and log out at any time.

2. Student Functions

2.1 Course Search and Viewing

- FR06: The system shall allow a student to view the list of available courses and open class sections for the selected semester.
- FR07: The system shall allow a student to search and filter class sections by course code, course name, lecturer, or schedule.
- FR08: The system shall display detailed information for each class section, including number of credits, lecturer name, room, day of the week, time slot, maximum capacity, and remaining available seats.
- FR09: The system shall display the prerequisite requirements for each course.

2.2. Course Registration and Cancel

- FR10: The system shall allow a student to select and register for multiple courses in a single registration period.
- FR11: The system shall verify that the student has completed all prerequisite courses before allowing registration for a course.
- FR12: The system shall check for schedule conflicts and prevent a student from registering for two class sections with overlapping study times.

- FR13: The system shall verify that the class section still has available seats before confirming the registration.
- FR14: The system shall check the total number of registered credits and ensure it stays within the minimum and maximum credit limits allowed per semester (e.g., 12 to 25 credits).
- FR15: The system shall store the registration record and immediately reduce the available seats of the class section by one upon successful registration.
- FR16: The system shall allow a student to cancel one or more registered class sections during the open registration period.

2.3. Schedule and Tuition Viewing

- FR17: The system shall allow a student to view the complete list of all courses they have successfully registered for in the semester.
- FR18: The system shall display the student's personal weekly class schedule (timetable) based on their registered courses.
- FR19: The system shall calculate and display the total number of registered credits and the total tuition fee for the student.
- FR20: The system shall allow a student to view their registration history, including registration timestamps and status (Registered, Canceled).

3. Lecturer Functions

- FR21: The system shall allow a lecturer to view all class sections assigned to them for the semester.
- FR22: The system shall allow a lecturer to view their personal teaching schedule (date, time, and room for each assigned class).
- FR23: The system shall allow a lecturer to view the list of enrolled students for each assigned class section.
- FR24: The system shall allow a lecturer to export or print the student list of their assigned classes.

4. Admin Functions

4.1 User Account Management

- FR25: The system shall allow the admin to create student accounts.
- FR26: The system shall allow the admin to create lecturer accounts.

- FR27: The system shall allow the admin to edit user account information.
- FR28: The system shall allow the admin to lock and unlock user accounts.

4.2. Course, Class, and Registration Period Management

- FR29: The system shall allow the admin to create, update, and delete courses (including course code, name, credits, and prerequisites).
- FR30: The system shall allow the admin to create class sections for a course in a specific semester and set the maximum student capacity, room, and schedule.
- FR31: The system shall allow the admin to assign a lecturer to a class section.
- FR32: The system shall allow the admin to configure the start date and end date for the course registration period.
- FR33: The system shall allow the admin to manually add or remove a student from a class section in special cases.
- FR34: The system shall allow the admin to cancel a class section if the number of enrolled students does not meet the minimum requirement.

4.3. Registration Reporting

- FR35: The system shall allow the admin to view registration statistics and reports by course and class section .
- FR36: The system shall allow the admin to view registration reports and total registered credits by student.

II. Non-Functional Requirements

1. Usability

- NFR01: The system shall provide a responsive web interface that displays properly on both desktop computers and mobile devices (screen widths from 360px to 1920px).
- NFR02: The system shall allow a student to search and register for a class section within a maximum of 3 clicks.
- NFR03: The system shall display clear and understandable error messages (e.g., "Schedule conflict", "Class is full", "Prerequisite not met") whenever a registration attempt fails.

2. Reliability

- NFR04: The system shall maintain at least 99% uptime during the official course registration period.
- NFR05: The system shall ensure that 0% probability of the number of registered students exceeding the class maximum capacity when multiple students register simultaneously.
- NFR06: The system shall automatically back up the database daily to prevent data loss in case of server failure.

3. Performance

- NFR07: The system shall load the course list and process a registration or drop request within 2 seconds under normal conditions and within 4 seconds during peak hours.
- NFR08: The system shall support at least 2,000 concurrent users accessing and registering for courses at the same time without system crashes.

4. Security

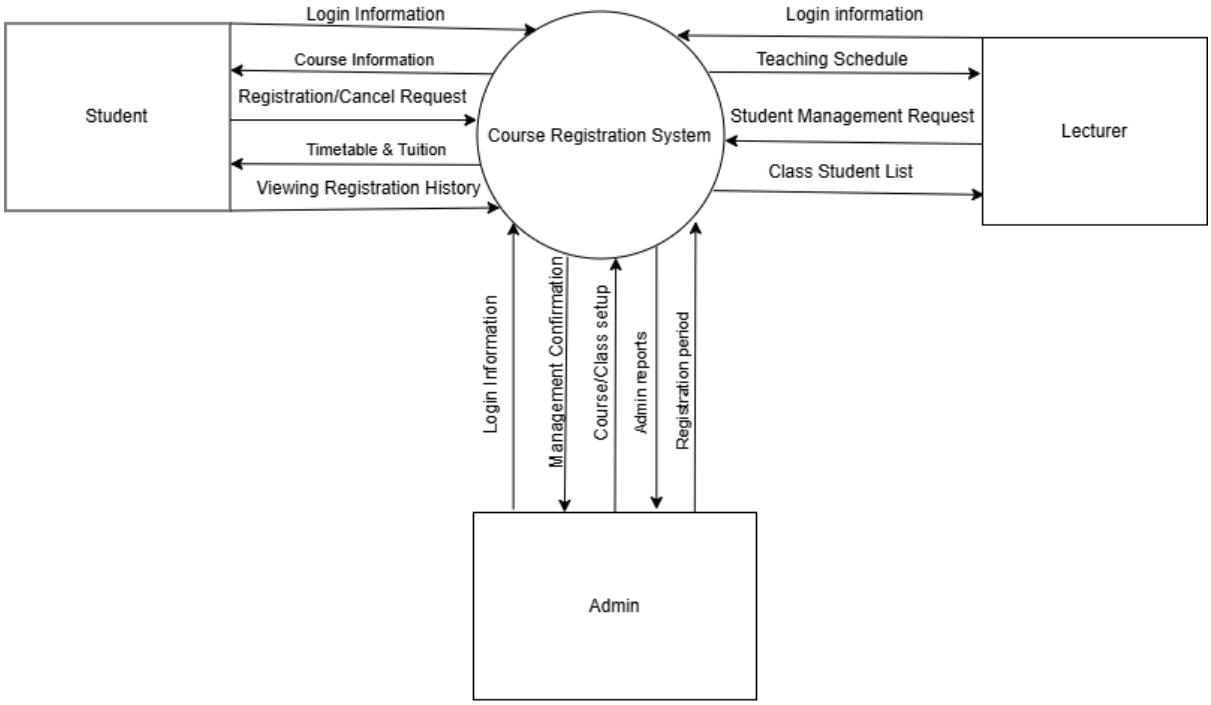
- NFR09: The system shall authenticate all users before granting access and encrypt all passwords in the database using hashing algorithms .
- NFR10: The system shall enforce role-based access control for Student, Lecturer, and Admin accounts.
- NFR11: The system shall automatically log out inactive user sessions after 15 minutes to prevent unauthorized access on shared computers.

5. Development Requirements

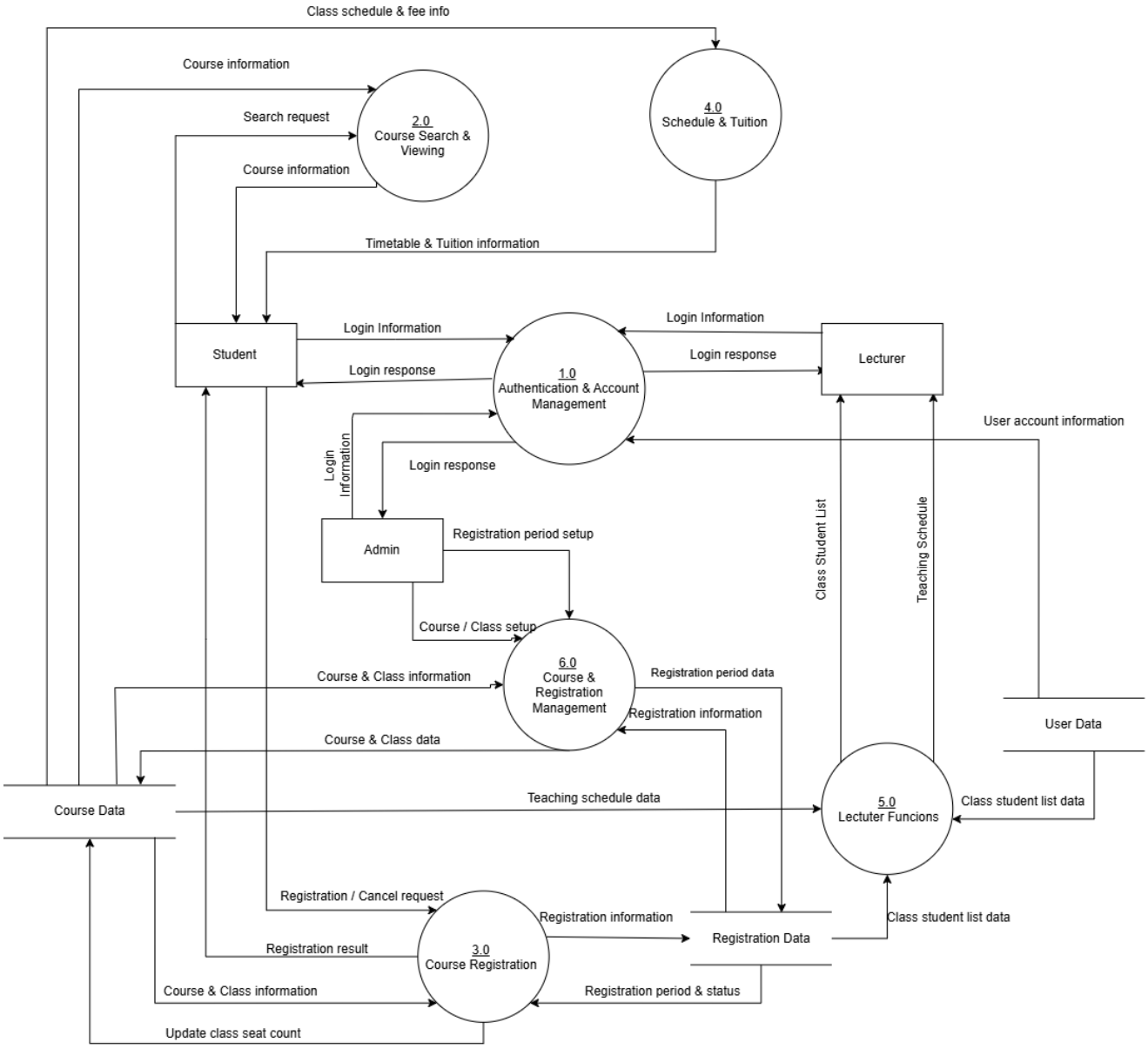
- NFR12: The system shall be developed using a modular architecture (MVC or 3-Tier architecture) to separate the user interface, business logic, and database layer.
- NFR13: The system shall use a relational database management system (MySQL or SQL Server) and manage source code via Git/GitHub.
- NFR14: Programming language: Python, HTML, CSS, JavaScript.

III. Data Flow Diagram

1. 0-level DFD

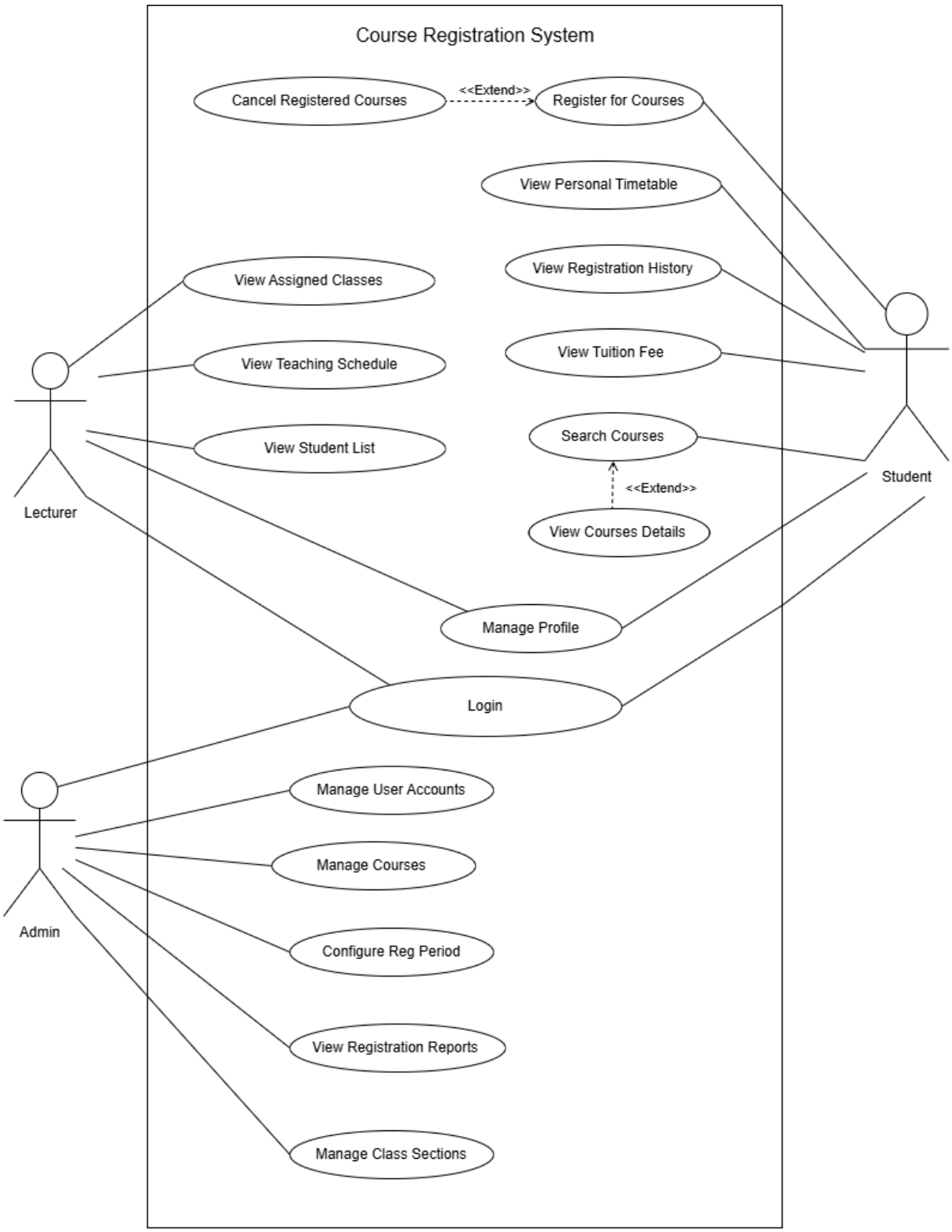


2. 1-level DFD



IV. Use case Diagram

1. Use Case Diagram Overview


Course Registration System
Cancel Registered Courses
<<Extend>>
Register for Courses
View Personal Timetable
View Registration History
View Tuition Fee
Search Courses
<<Extend>>
View Courses Details
View Assigned Classes
View Teaching Schedule
View Student List
Lecturer
Student
Manage Profile
Login
Admin
Manage User Accounts
Manage Courses
Configure Reg Period
View Registration Reports
Manage Class Sections

2. List of Actor

No.	Actor	Description
1	Student	A student who logs into the system to search and register for courses, view personal timetable, registration history, and tuition fee
2	Lecturer	A lecturer who is responsible for teaching classes, viewing assigned classes, teaching schedule, and student lists
3	Admin	An administrator who manages user accounts, courses, class sections, registration periods, and registration reports of the system

3. List of Use Case

No.	Use-case	Description	Group
1	Login	Log in to the system	Common
2	Manage Profile	View and update personal profile information	Common
3	Search Courses	Search for available courses by keyword or criteria	Student
4	View Course Details	View detailed information of a selected course	Student

No.	Use-case	Description	Group
5	Register for Courses	Register for courses during the registration period	Student
6	Cancel Registered Courses	Cancel courses that the student has registered	Student
7	View Personal Timetable	View the personal class timetable	Student
8	View Registration History	View the history of course registrations	Student
9	View Tuition Fee	View the tuition fee based on registered courses	Student
10	View Assigned Classes	View the list of classes assigned to the lecturer	Lecturer
11	View Teaching Schedule	View the teaching schedule of the lecturer	Lecturer
12	View Student List	View the list of students in each assigned class	Lecturer
13	Manage User Accounts	Create, update, lock/unlock, and delete user accounts	Admin
14	Manage Courses	Create, update, and delete courses in the system	Admin
15	Configure Registration Period	Set the open and close time of the course registration period	Admin

No.	Use-case	Description	Group
16	View Registration Reports	View statistics and reports of course registrations	Admin
17	Manage Class Sections	Create, update, and delete class sections of courses	Admin

4. Use Case: Login

4.1 Summary

Allows users (Student, Lecturer, Admin) to authenticate and access the system based on their roles.

4.2 Event Flow

4.2.1 Main Flow of Events

The user selects the Login function.

The system displays the login form.

The user enters a username (Student ID / Lecturer ID / Admin ID) and password. The system verifies the credentials and identifies the user's role.

The system grants access and redirects the user to the corresponding dashboard based on their role.

4.2.2 Alternative Flows

If the username or password is incorrect, the system refuses access, displays an error message, and requests re-entry.

If the user's account is locked or suspended, the system blocks the login attempt and displays an account locked notification.

4.3 Special Requirements

Passwords must be encrypted in the database using hashing algorithms.

The system must automatically log out inactive user sessions after 15 minutes.

4.4 Pre-condition

The user has a valid account in the system.

4.5 Post-condition

The user is logged into the system successfully and redirected to their role-specific dashboard.

4.6 Extension Points

None.

5. Use Case: Manage Profile

5.1 Summary

This use-case allows any authenticated user (Student, Lecturer, or Admin) to view and update their personal profile details.

5.2 Event Flow

5.2.1 Main Flow of Events

The user selects the Manage Profile function.

The system displays the user's profile information and password change options. The user selects the change password option, enters the current password, new password, and confirmation password.

The system validates the input and updates the password in the database.

The system displays a confirmation message.

5.2.2 Alternative Flows

If the current password is incorrect or the new password confirmation does not match, the system displays an error message and requests re-entry.

5.3 Special Requirements

Primary keys like Email or Student ID cannot be modified. Password updates must be securely hashed (BCrypt/Argon2).

5.4 Pre-condition

The user is logged into the system.

5.5 Post-condition

If successful, profile details are updated in the database. Otherwise, the system state remains unchanged.

5.6 Extension Points

None.

6. Use Case: Search Courses

6.1 Summary

Allows students to view, search, and filter available courses and open class sections for a selected semester.

6.2 Event Flow

6.2.1 Main Flow of Events

The student selects the Search Courses function and chooses a semester.

The system loads and displays the list of available courses and open class sections for the selected semester.

The student enters search criteria or applies filters by course code, course name, lecturer, or schedule.

The system queries the database and displays the matching class sections.

6.2.2 Alternative Flows

If no class sections match the search criteria, the system displays a "No matching courses found" message.

6.3 Special Requirements

The system must load the course list within 2 seconds under normal conditions and within 4 seconds during peak hours.

6.4 Pre-condition

The student is logged into the system.

6.5 Post-condition

The list of available and filtered class sections is displayed to the student.

6.6 Extension Points

View Courses Details.

7. Use Case: View Courses Details

7.1 Summary

Allows students to view detailed information and prerequisite requirements of a specific class section (extends Search Courses).

7.2 Event Flow

7.2.1 Main Flow of Events

From the course search results, the student selects a specific class section to view its details.

The system retrieves the detailed information of the selected class section.

The system displays the number of credits, lecturer name, room, day of the week, time slot, maximum capacity, remaining available seats, and prerequisite requirements.

7.2.2 Alternative Flows

None.

7.3 Special Requirements

None.

7.4 Pre-condition

The student is viewing the list of courses in the Search Courses function.

7.5 Post-condition

The detailed information of the selected class section is displayed.

7.6 Extension Points

None.

8. Use Case: Register for Courses

8.1 Summary

Allows students to select and register for one or multiple class sections during an active registration period.

8.2 Event Flow

8.2.1 Main Flow of Events

The student selects the Register for Courses function.

The student selects one or multiple open class sections and submits the registration request.

The system verifies that the student has completed all prerequisite courses for the selected courses.

The system checks that there are no schedule conflicts (overlapping study times or exam dates).

The system verifies that the selected class sections still have available seats.

The system checks that the student's total registered credits remain within the allowed semester limits (e.g., 12 to 25 credits).

The system stores the registration record, immediately reduces the available seats of each registered class section by one, and displays a success message.

8.2.2 Alternative Flows

If any prerequisite is not met, the system rejects the registration for that course and displays the error message "Prerequisite not met".

If a schedule or exam conflict is detected, the system prevents registration and displays the error message "Schedule conflict".

If a class section has no remaining seats, the system prevents registration and displays the error message "Class is full".

If the total credits fall outside the allowed limit, the system displays a credit limit warning and prevents confirmation.

8.3 Special Requirements

The student must be able to search and register for a class section within a maximum of 3 clicks.

The system must process a registration request within 2 seconds under normal conditions (4 seconds during peak hours) and ensure 0% probability of over-enrollment when multiple students register simultaneously.

8.4 Pre-condition

The student is logged in, and the current date/time is within the active registration period configured by the Admin.

8.5 Post-condition

The registration record is saved, the student is enrolled in the selected class sections, and the available seats are decremented by one.

8.6 Extension Points

None

9. Use Case: Cancel Registered Courses

9.1 Summary

Allows students to cancel one or more registered class sections during the open registration period .

9.2 Event Flow

9.2.1 Main Flow of Events

While viewing their registered courses during the registration period, the student selects one or more registered class sections and chooses the Cancel .

The system prompts the student for confirmation.

The student confirms the cancellation.

The system checks that the remaining credits still comply with semester rules, updates the registration status to " Canceled ", increases the available seats of the class section by one, and displays a confirmation message.

9.2.2 Alternative Flows

If the registration period has ended, the system blocks the cancellation request and displays an error message.

If the student cancels the confirmation prompt, the system aborts the operation and retains the registration.

9.3 Special Requirements

The drop request must be processed within 2 seconds under normal conditions and within 4 seconds during peak hours.

9.4 Pre-condition

The student is logged in, has at least one registered course, and the registration period is currently open.

9.5 Post-condition

The selected class section is removed from the student's active schedule, the action is logged in registration history as " Canceled ", and the class section's available seats increase by one.

9.6 Extension Points

None.

10. Use Case: View Personal Timetable

10.1 Summary

Allows students to view their complete list of registered courses and their personal weekly class schedule for the semester.

10.2 Event Flow

10.2.1 Main Flow of Events

The student selects the View Personal Timetable function and chooses a semester. The system retrieves all courses the student has successfully registered for in that semester.

The system generates and displays the student's personal weekly timetable showing course names, days of the week, time slots, rooms, and lecturers.

10.2.2 Alternative Flows

If the student has not registered for any courses in the selected semester, the system displays an empty timetable with a notification message.

10.3 Special Requirements

The timetable interface must be responsive and display properly on both desktop and mobile screens .

10.4 Pre-condition

The student is logged into the system.

10.5 Post-condition

The student's weekly class schedule is displayed.

10.6 Extension Points

None.

11. Use Case: View Registration History

11.1 Summary

Allows students to view the history of all their course registration and cancellation actions.

11.2 Event Flow 11.2.1 Main Flow of Events

The student selects the View Registration History function.

The system retrieves the log of all registration transactions made by the student.

The system displays the registration history list, including course code, class section, registration timestamps, and status ("Registered" or " Canceled ").

11.2.2 Alternative Flows

If no registration actions have been performed, the system displays a "No registration history available" message.

11.3 Special Requirements

None.

11.4 Pre-condition

The student is logged into the system.

11.5 Post-condition

The student's registration history is displayed accurately.

11.6 Extension Points

None.

12. Use Case: View Tuition Fee

12.1 Summary

Allows students to view their total registered credits and the calculated tuition fee for the semester.

12.2 Event Flow 12.2.1 Main Flow of Events

The student selects the View Tuition Fee function and selects a semester.

The system retrieves the list of courses the student has successfully registered for in the semester.

The system calculates the total number of registered credits and the corresponding total tuition fee.

The system displays the itemized course list, credit count per course, total registered credits, and total tuition fee.

12.2.2 Alternative Flows

If the student has no registered courses, the system displays 0 credits and a total tuition fee of 0.

12.3 Special Requirements

None.

12.4 Pre-condition

The student is logged into the system.

12.5 Post-condition

The total registered credits and total tuition fee are displayed to the student.

12.6 Extension Points

None.

13. Use Case: View Assigned Classes

13.1 Summary

Allows a Lecturer to view all class sections assigned to them for a semester.

13.2 Event Flow

13.2.1 Main Flow of Events

The Lecturer selects the View Assigned Classes function.

The system retrieves the Lecturer's ID from the current session.

The system displays the list of class sections assigned to the Lecturer for the current semester, including class code, course name, number of credits, room, and number of enrolled students.

The Lecturer may select another semester. The system reloads and displays the assigned class sections of that semester.

13.2.2 Alternative Flows

If no class section is assigned to the Lecturer in the selected semester, the system displays a "No assigned classes found" notification.

If the data cannot be retrieved, the system displays an error message and asks the Lecturer to try again.

If the session has expired, the system redirects the user to the Login page.

13.3 Special Requirements

The list must be loaded within 2 seconds under normal conditions.

A Lecturer can only view class sections assigned to themselves.

13.4 Pre-condition

The user is logged in with the Lecturer role.

13.5 Post-condition

The list of assigned class sections is displayed to the Lecturer. No data is changed.

13.6 Extension Points

None.

14. Use Case: View Teaching Schedule

14.1 Summary

Allows lecturers to view all class sections assigned to them and their personal teaching schedule for the semester.

14.2 Event Flow

14.2.1 Main Flow of Events

The lecturer selects the View Teaching Schedule function and chooses a semester. The system retrieves all class sections assigned to the lecturer for the selected semester.

The system displays the assigned class sections along with the personal teaching schedule (date, day of the week, time slot, and room for each class).

14.2.2 Alternative Flows

If no class sections are assigned to the lecturer for the semester, the system displays a "No assigned classes found" message.

14.3 Special Requirements

None.

14.4 Pre-condition

The lecturer is logged into the system.

14.5 Post-condition

The lecturer's assigned classes and teaching schedule are displayed.

14.6 Extension Points

None.

15. Use Case: View Student List

15.1 Summary

Allows lecturers to view, export, or print the list of enrolled students for each of their assigned class sections.

15.2 Event Flow

15.2.1 Main Flow of Events

The lecturer selects the View Student List function and chooses an assigned class section.

The system retrieves and displays the list of enrolled students for that class section (including Student ID, full name, and contact details).

The lecturer selects the Export or Print option.

The system generates a downloadable file or opens the print view of the student list.

15.2.2 Alternative Flows

If no students are currently enrolled in the selected class section, the system displays an empty list message and disables the export/print buttons.

15.3 Special Requirements

None.

15.4 Pre-condition

The lecturer is logged into the system and has at least one assigned class section.

15.5 Post-condition

The enrolled student list is displayed and optionally exported or printed.

15.6 Extension Points

None.

16. Use Case: Manage User Accounts

16.1 Summary

Allows the Admin to create, update, lock/unlock, and delete user accounts.

16.2 Event Flow

16.2.1 Main Flow of Events

The Admin selects the Manage User Accounts function.

The system displays the current list of user accounts.

The Admin selects an action (create, update, lock/unlock, or delete) and enters the required

information.

The system validates the input and updates the user account records in the database.

The system displays a success message.

16.2.2 Alternative Flows

If the Admin enters invalid data (e.g., duplicate ID), the system displays an error message and

requests re-entry.

If the Admin attempts to delete an account with active dependencies, the system blocks the

action and suggests locking the account instead.

16.3 Special Requirements

None.

16.4 Pre-condition

The Admin is logged into the system.

16.5 Post-condition

The user account is successfully created, updated, locked/unlocked, or deleted.

16.6 Extension Points

None.

17. Use Case: Configure Registration Period

17.1 Summary

Allows the administrator to configure the start date and end date for the course registration period of a semester.

17.2 Event Flow

17.2.1 Main Flow of Events

The admin selects the Configure Reg Period function.

The system displays the current registration period settings for the semester.

The admin enters or modifies the start date/time and end date/time for the course registration period and clicks Save.

The system validates the dates and stores the configuration.

The system displays a confirmation message.

17.2.2 Alternative Flows

If the end date/time is earlier than or equal to the start date/time, the system displays an invalid date range error and requests re-entry.

17.3 Special Requirements

None.

17.4 Pre-condition

The admin is logged into the system.

17.5 Post-condition

The course registration period is configured and enforced across the system for students.

17.6 Extension Points

None.

18. Use Case: Manage Courses

18.1 Summary

Allows the administrator to create, update, and delete courses, as well as manually add or remove students from class sections in special cases.

18.2 Event Flow

18.2.1 Main Flow of Events

The admin selects the Manage Courses function.

The system displays the list of existing courses and management options.

The admin chooses to create a new course, update an existing course, or delete a course (specifying course code, name, credits, and prerequisites), or manually add/remove a student in a class section.

The system validates the input data and executes the requested update in the database.

The system displays a success message and refreshes the course list.

18.2.2 Alternative Flows

If the admin attempts to create a course with an existing course code, the system displays a duplicate code error. If the admin attempts to delete a course that currently has active class sections with enrolled students, the system blocks the deletion and displays a warning message.

18.3 Special Requirements

None.

18.4 Pre-condition

The admin is logged into the system.

18.5 Post-condition

The course is successfully created, modified, or deleted in the system.

18.6 Extension Points

None

19. Use Case: View Registration Reports

19.1 Summary

Allows the Admin to view statistics and reports of course registrations.

19.2 Event Flow

19.2.1 Main Flow of Events

The Admin selects the View Registration Reports function.

The system provides filter options (e.g., by semester or department).

The Admin selects the desired filters and requests the report.

The system queries the database and displays the statistical registration data.

19.2.2 Alternative Flows

If no registration data matches the selected filters, the system indicates that no data was found.

19.3 Special Requirements

Reports should be available for export.

19.4 Pre-condition

The Admin is logged into the system.

19.5 Post-condition

The Admin successfully views the registration statistics.

19.6 Extension Points

None.

20. Use Case: Manage Class Sections

20.1 Summary

Allows the Admin to create, update, and delete class sections of courses.

20.2 Event Flow

20.2.1 Main Flow of Events

The Admin selects the Manage Class Sections function.

The system displays a list of existing class sections.

The Admin inputs data to create, update, or delete a specific class section.

The system validates scheduling, room capacity, and lecturer availability.

The system saves the changes and displays a success prompt.

20.2.2 Alternative Flows

If a scheduling conflict is detected for a lecturer or room, the system blocks the creation/update and displays a conflict warning.

20.3 Special Requirements

None.

20.4 Pre-condition

The Admin is logged into the system.

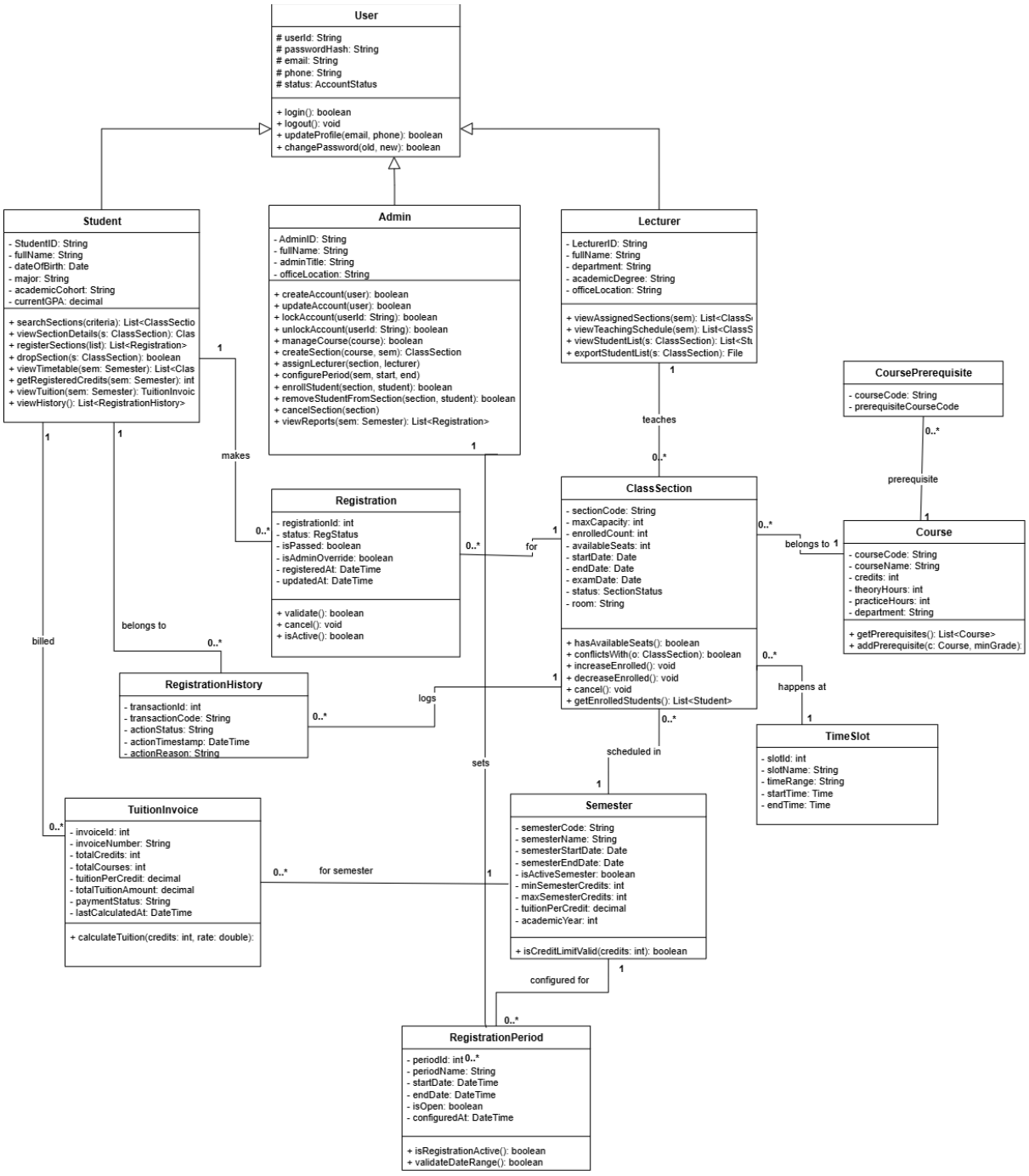
20.5 Post-condition

The class section is successfully created, updated, or deleted.

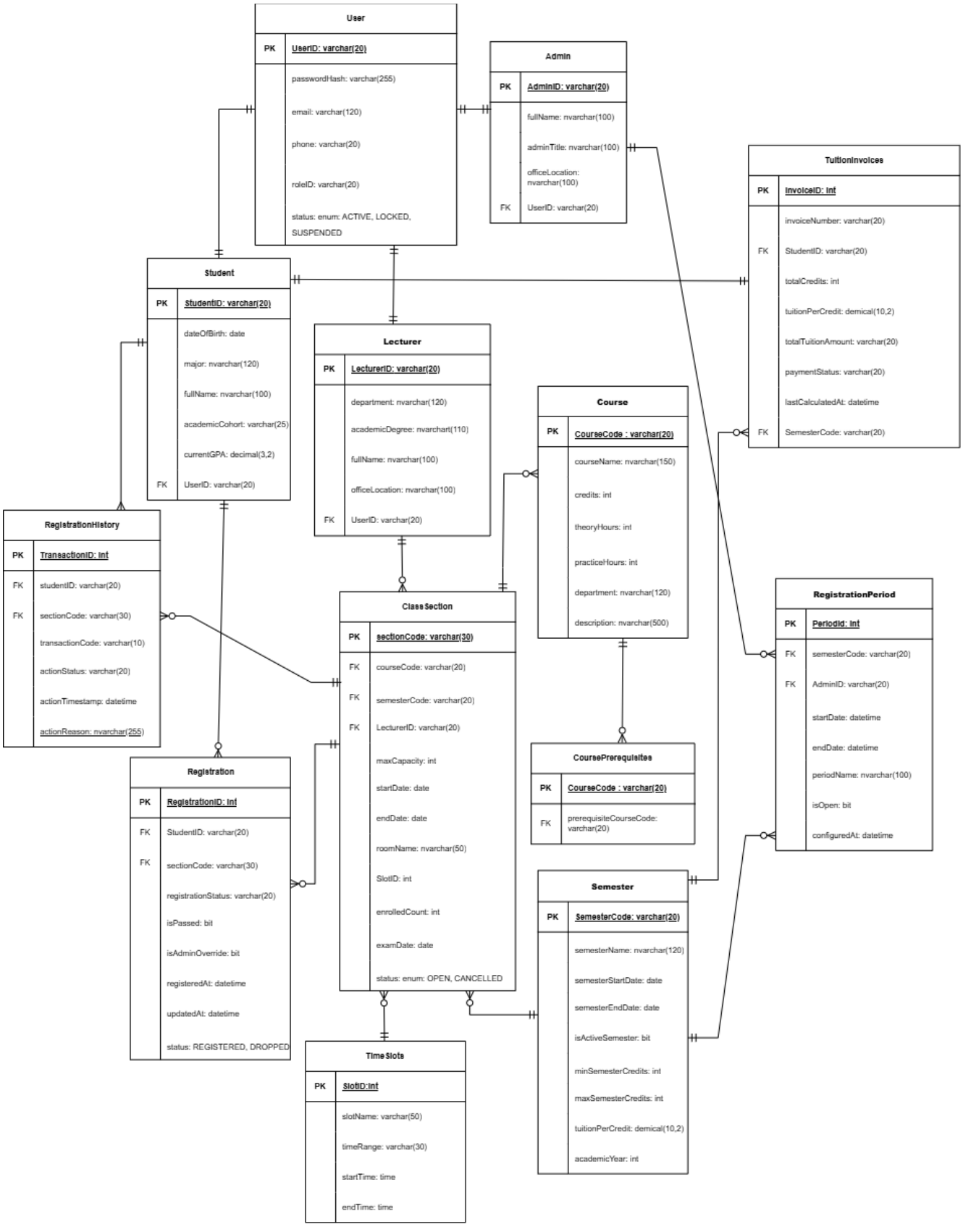
20.6 Extension Points

None.

V. Class Diagram

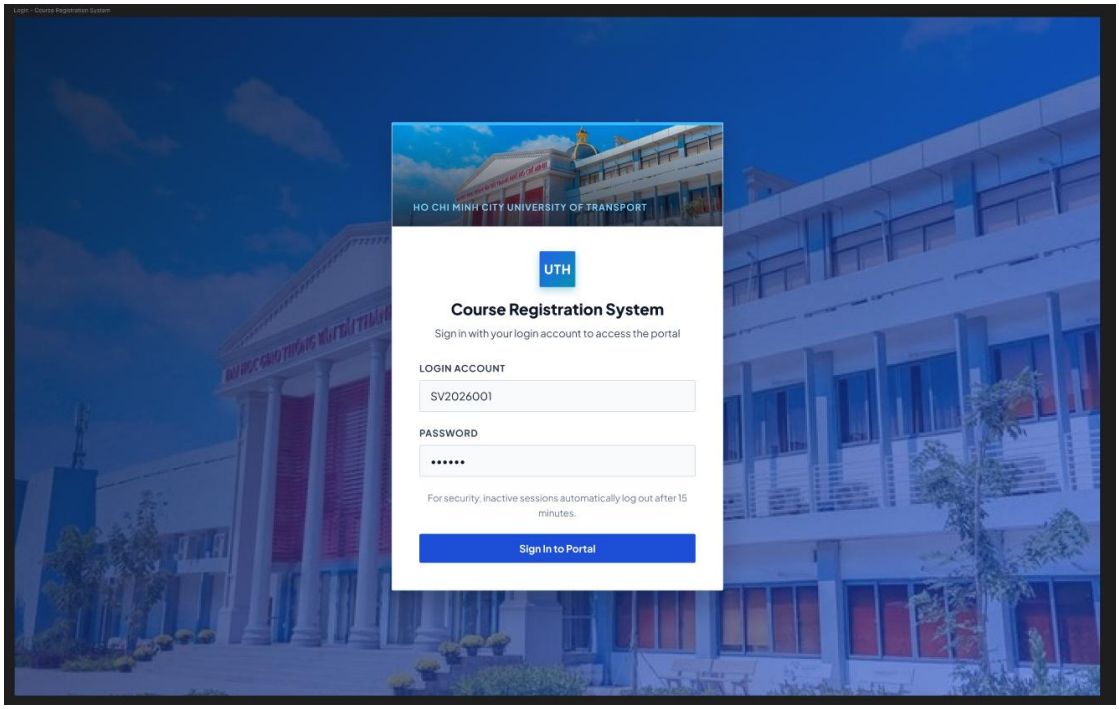


VI. Data Model



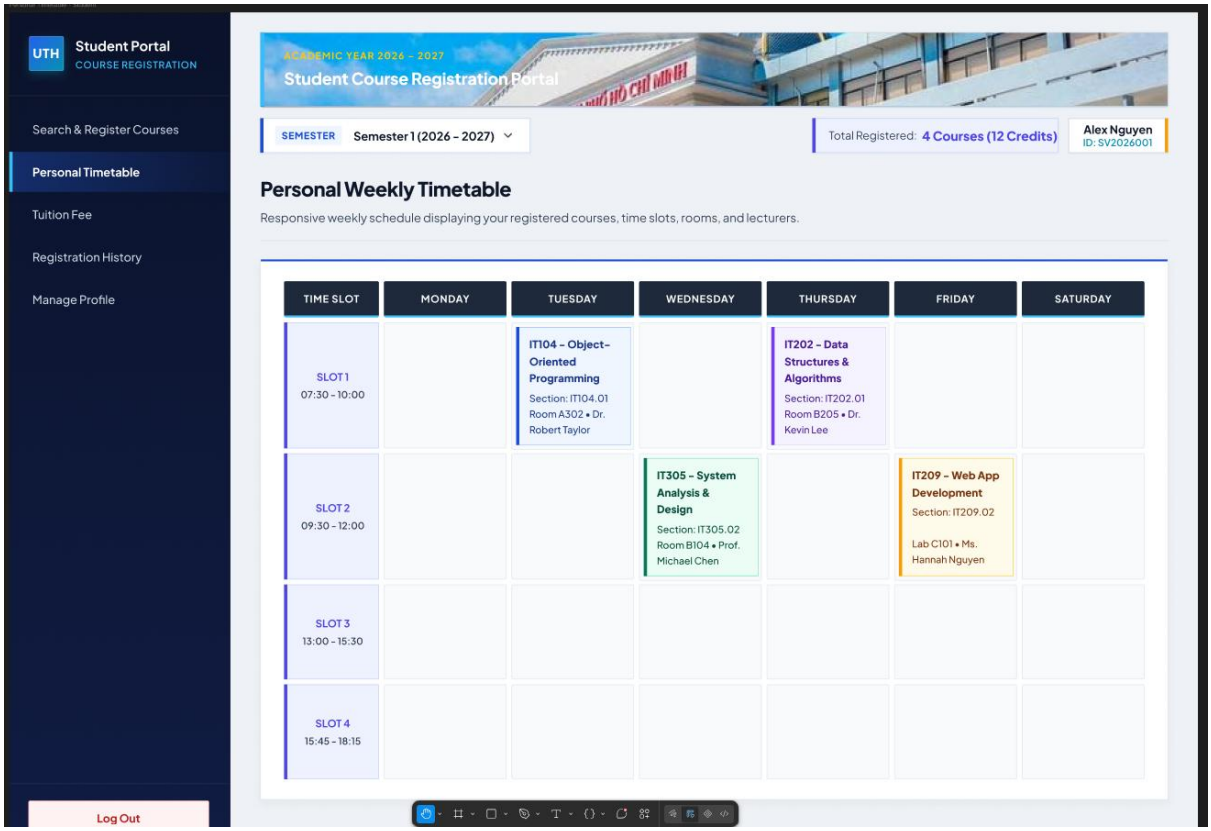
VII. Interface Design Description

Login

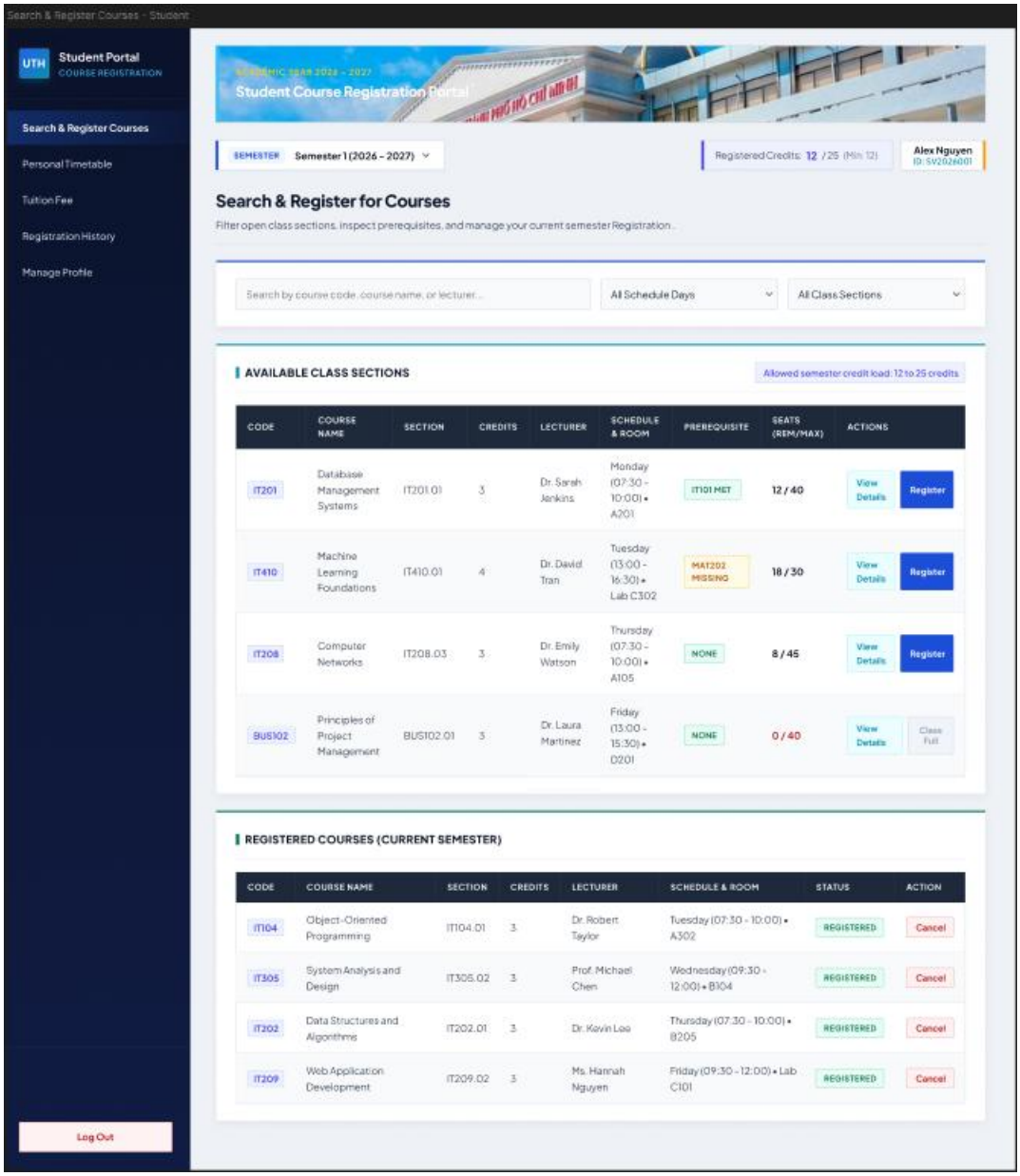


1. Student

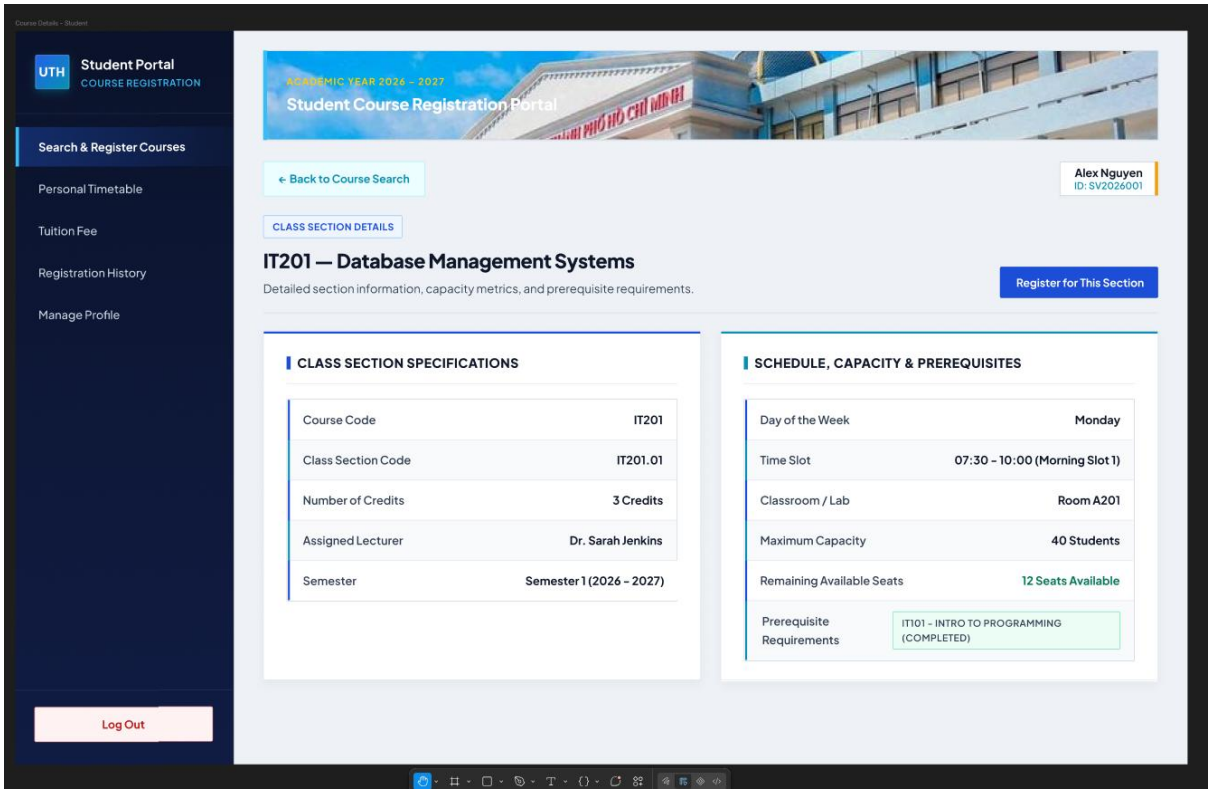
Personal Timetable



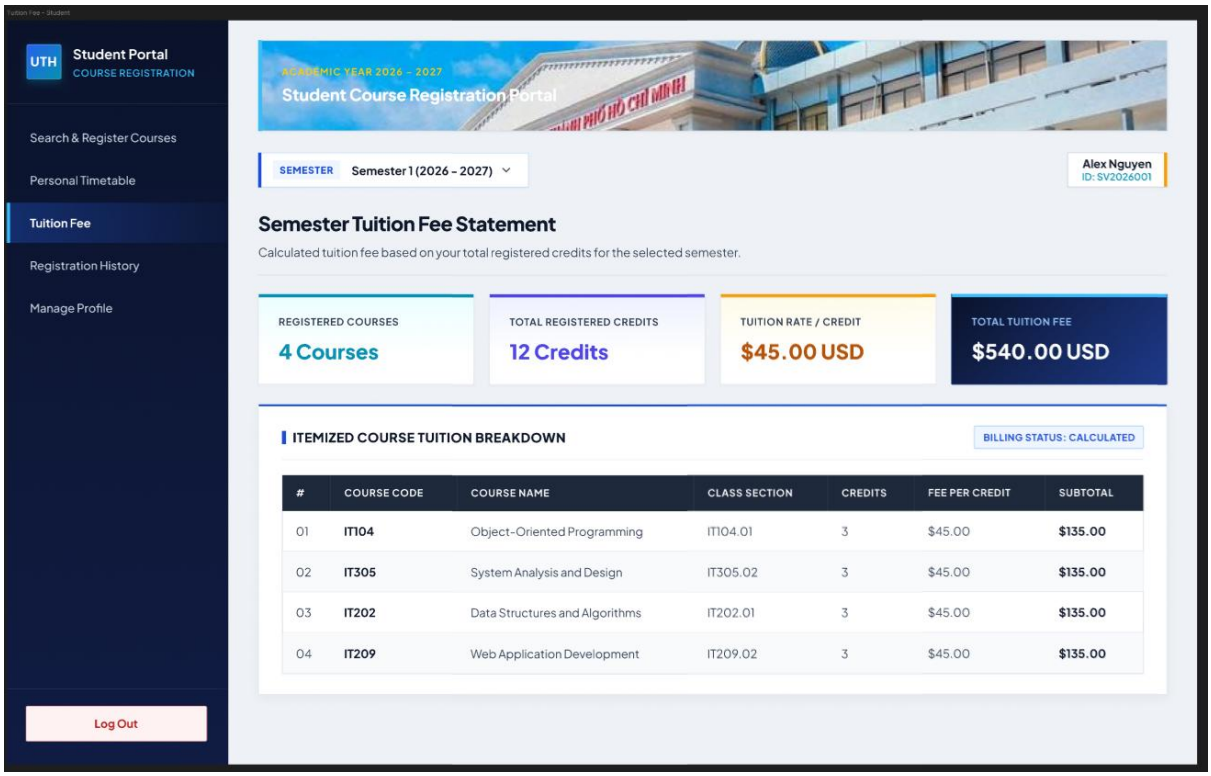
Search & Register



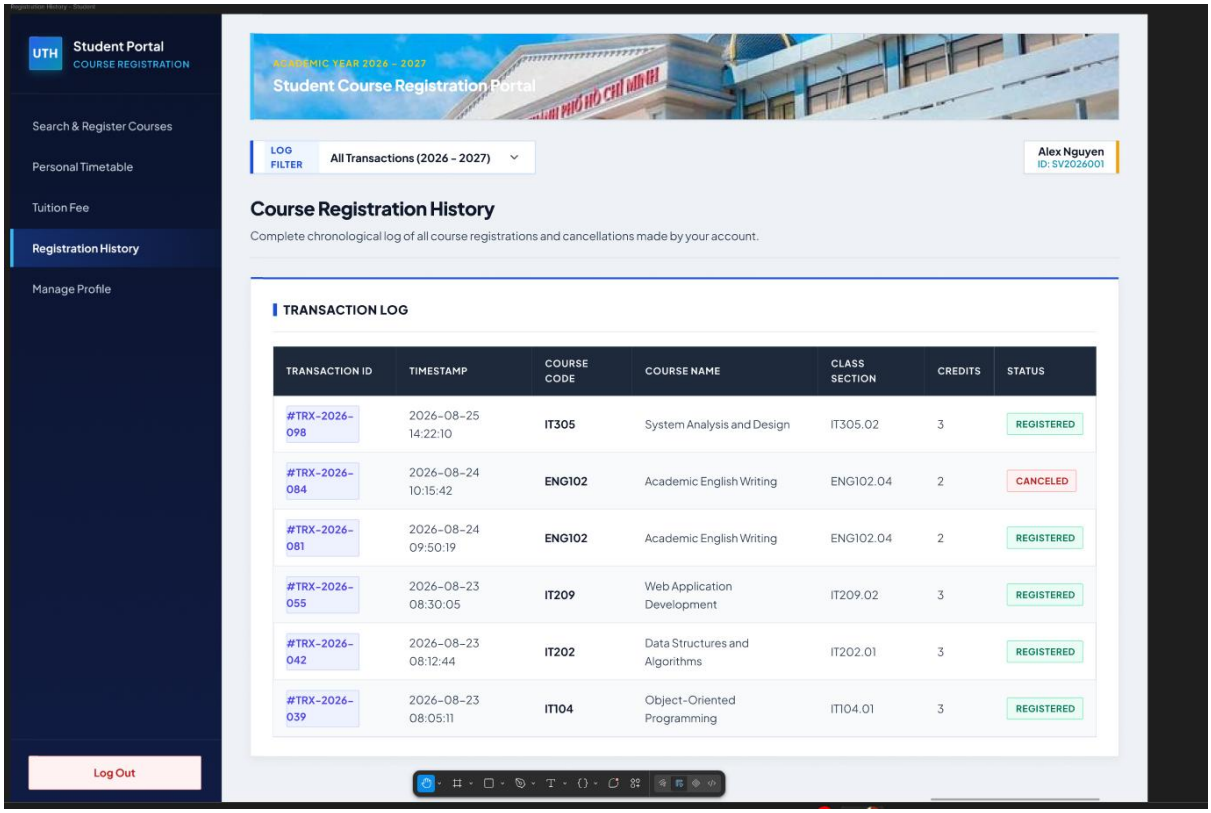
Course Detail



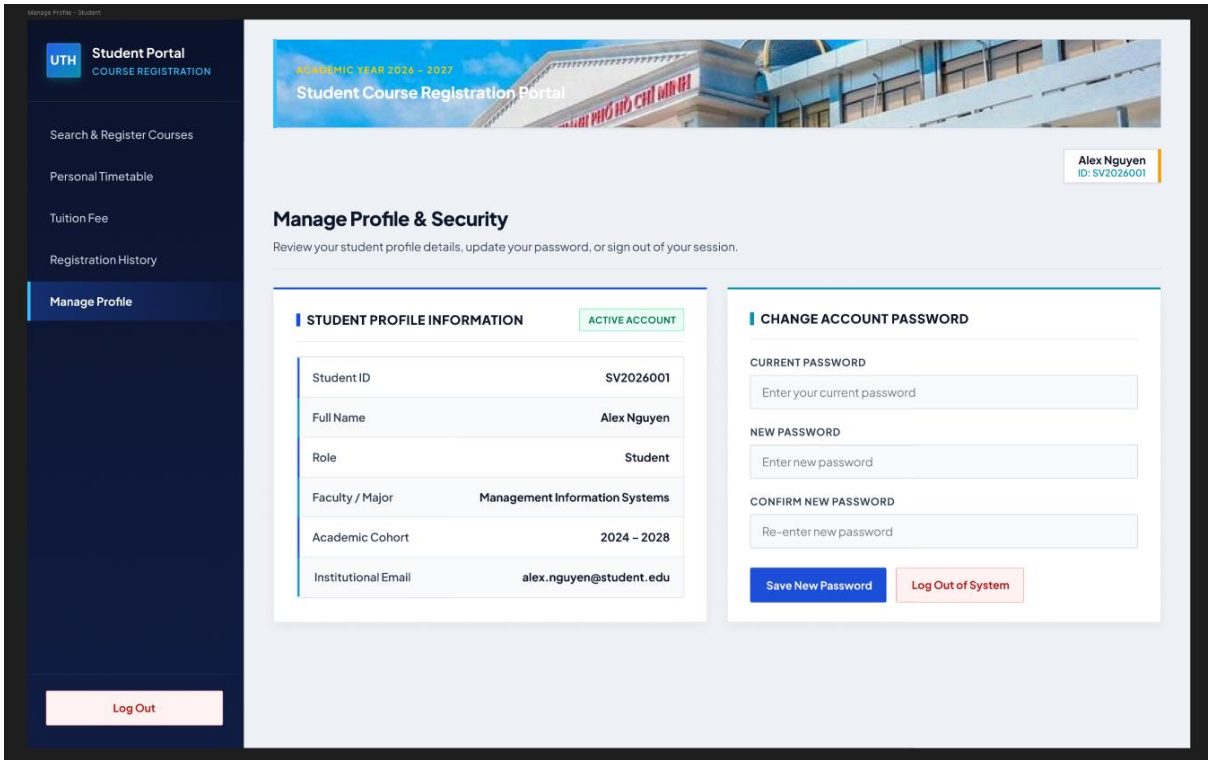
Tuition Fee



Registration History



Manage Profile

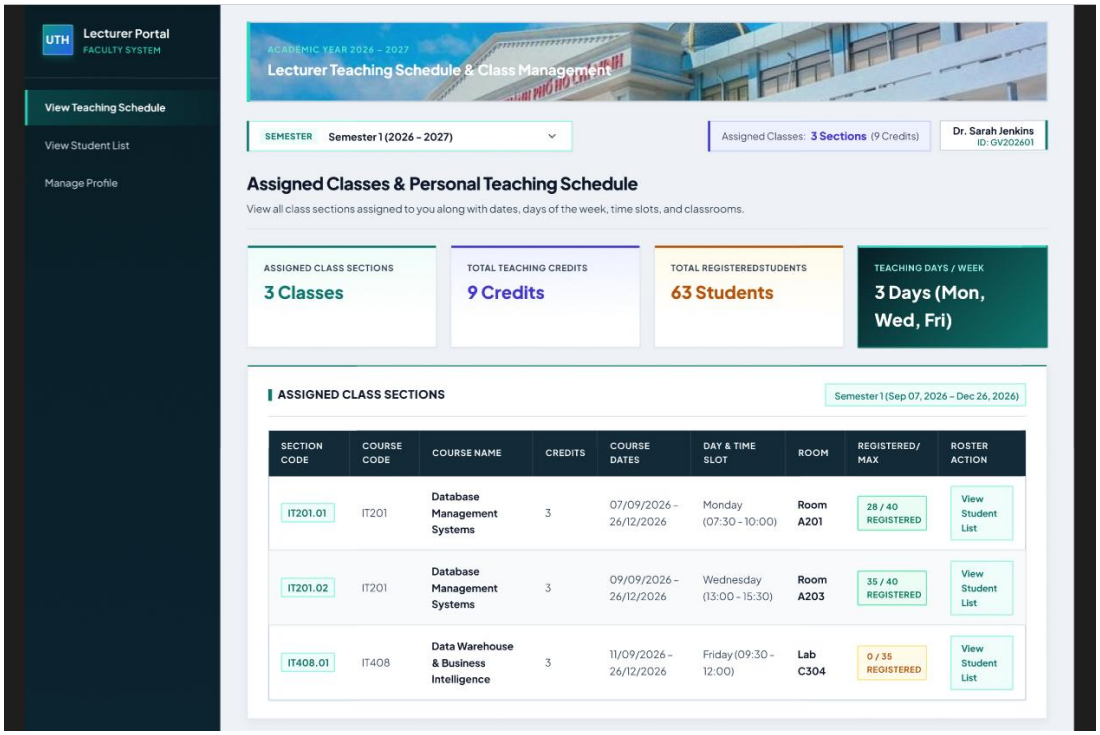


2. Lecturer

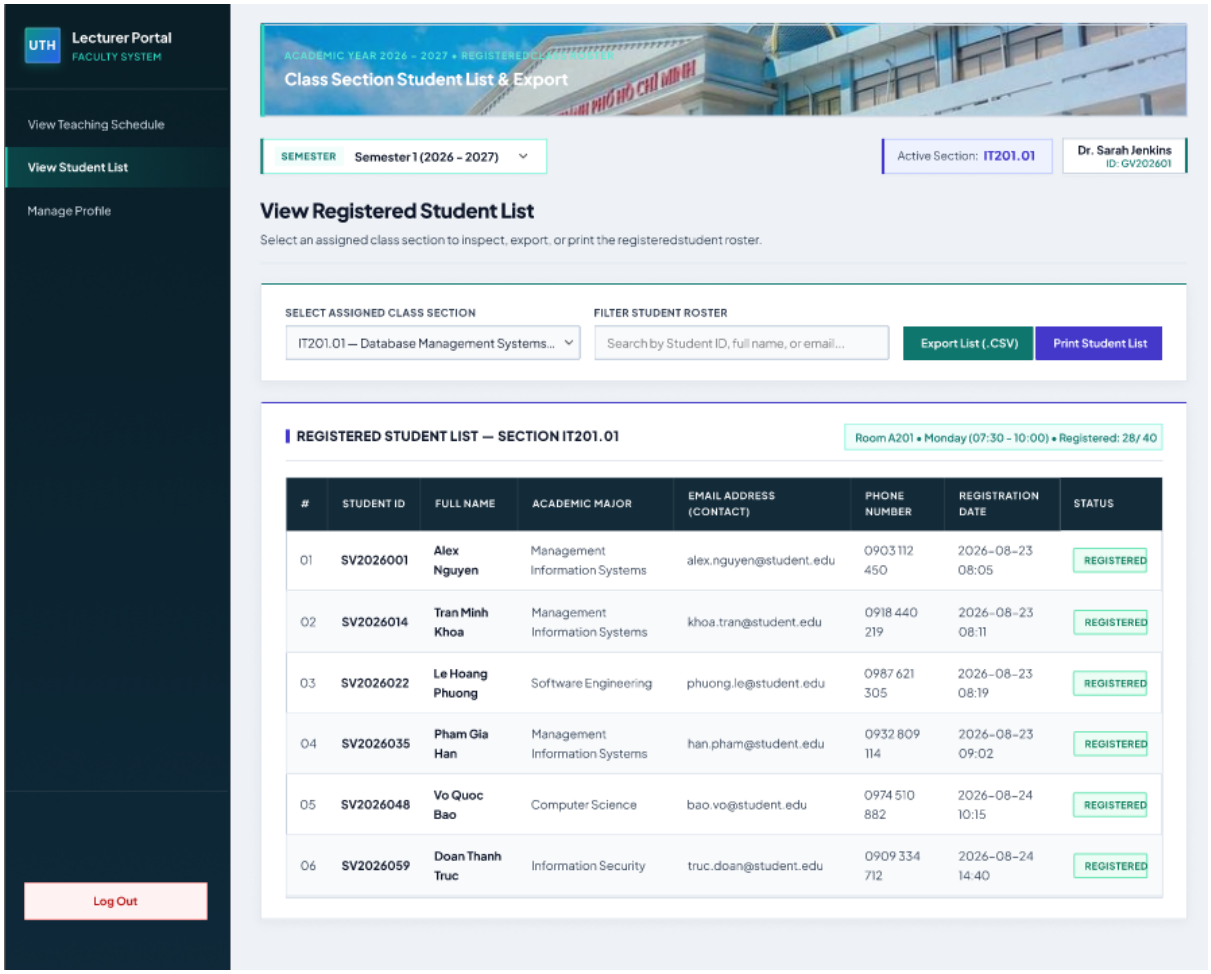
View Teaching Schedule



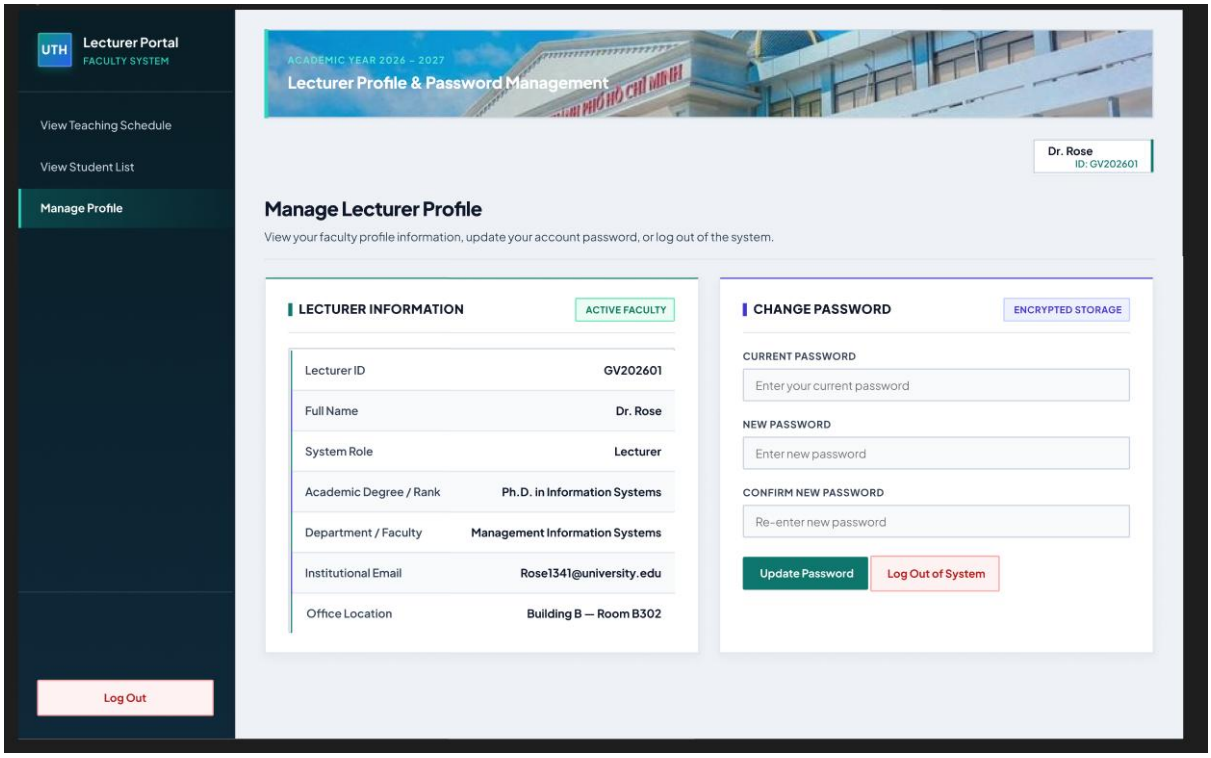
View Assigned Classes



View Student List

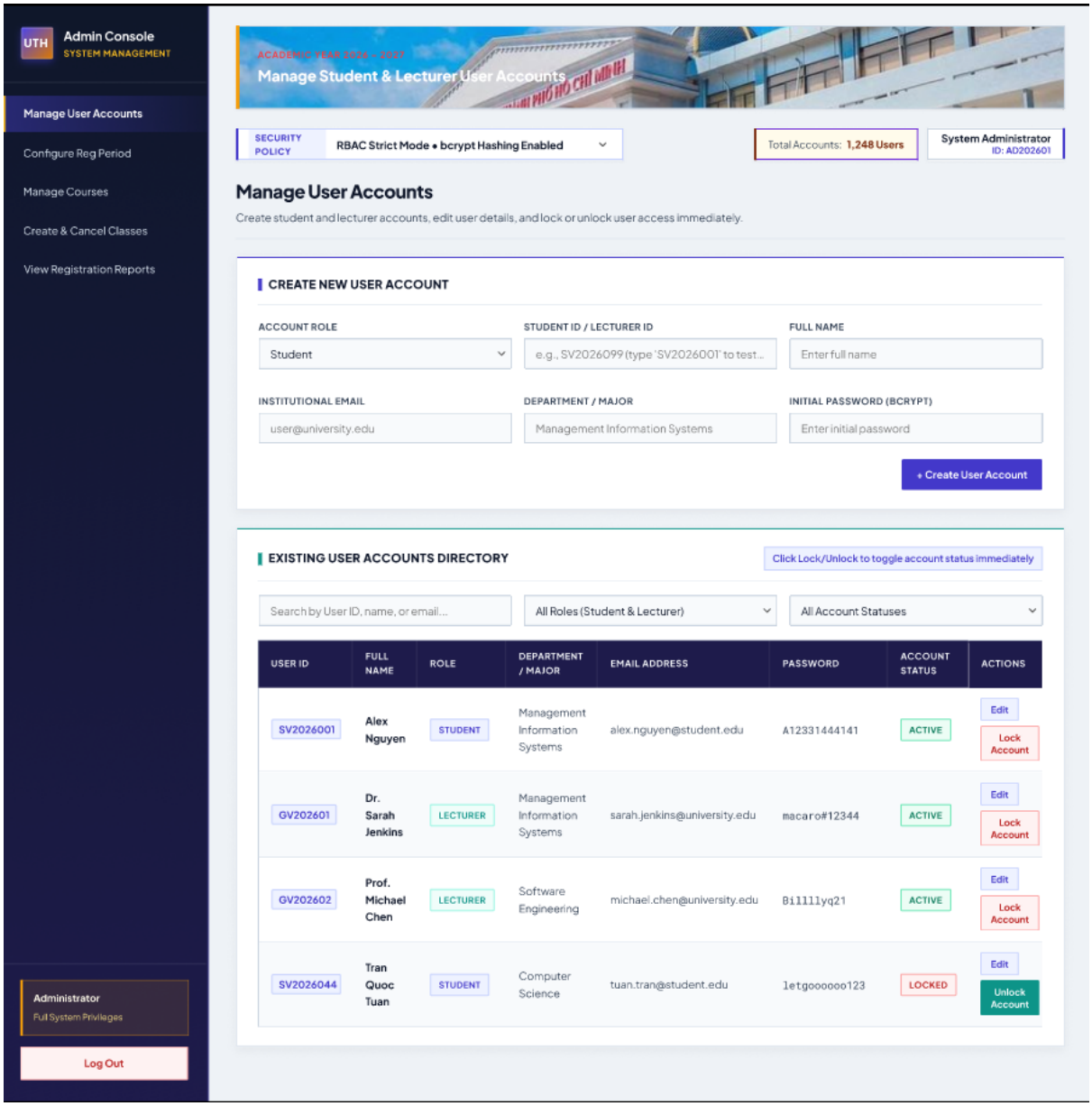


Manage Profile

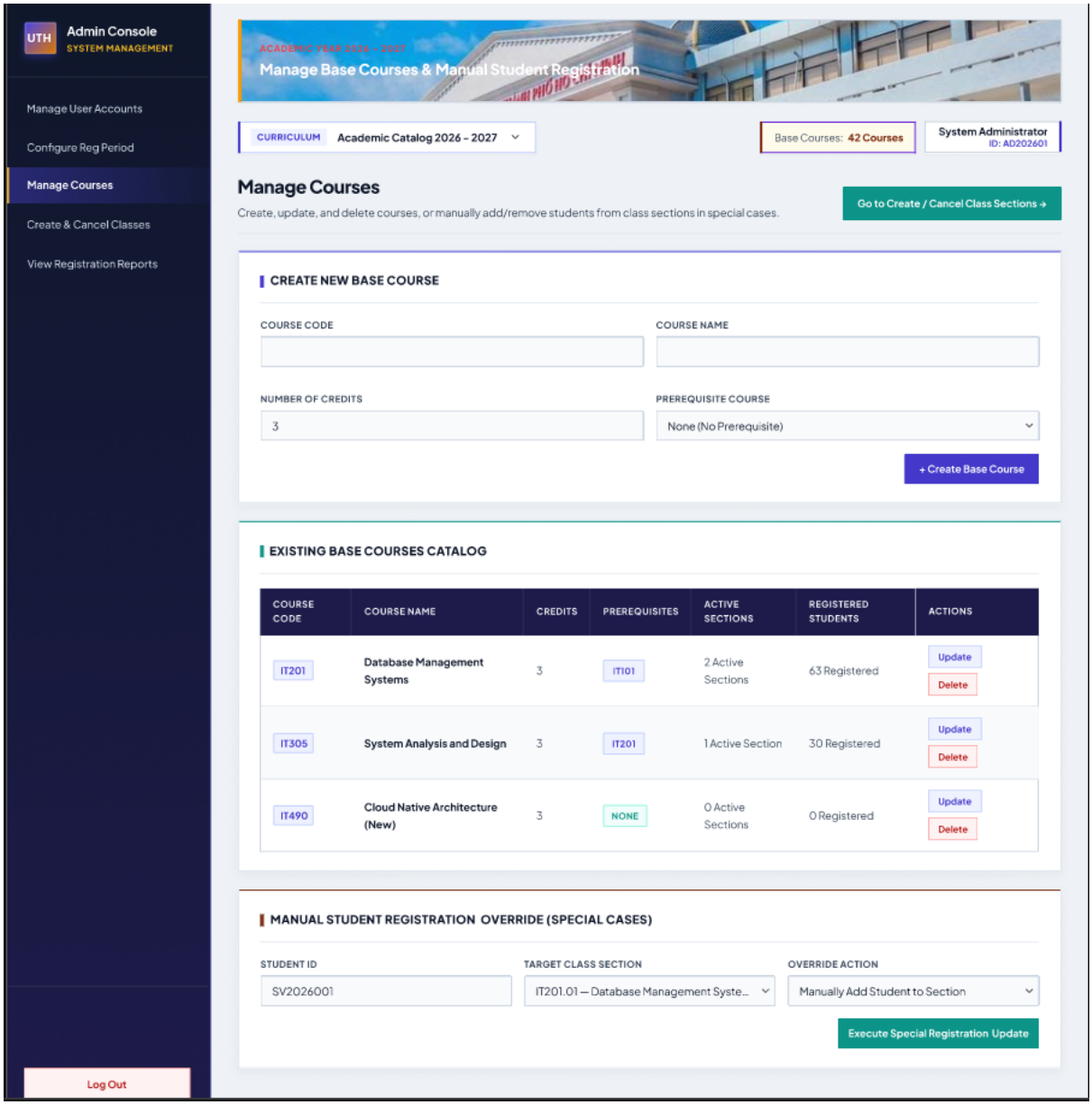


3. Admin

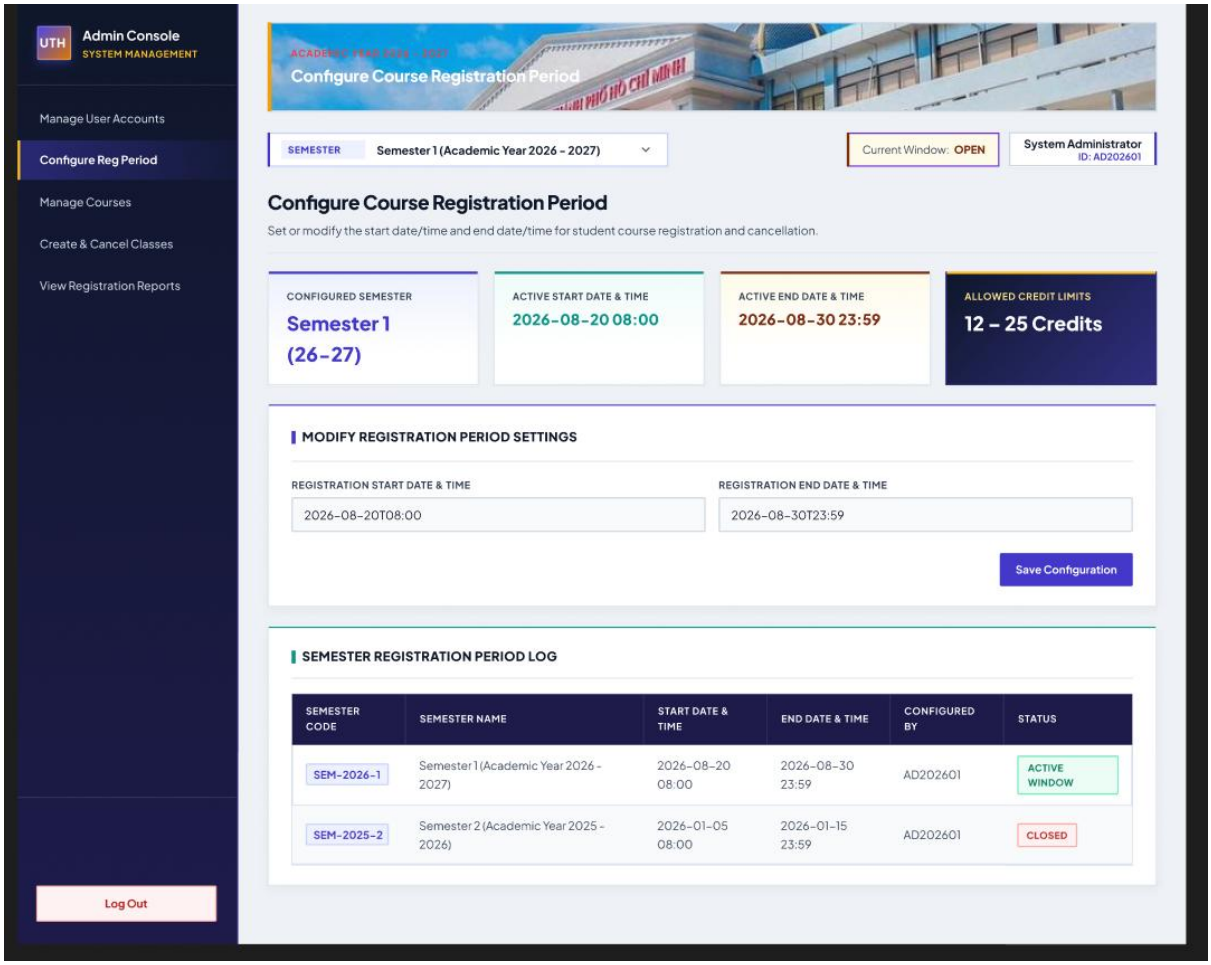
Manage User Accounts


UTH
Admin Console
SYSTEM MANAGEMENT
Manage User Accounts
Configure Reg Period
Manage Courses
Create & Cancel Classes
View Registration Reports
Administrator
Full System Privileges
Log Out
ACADEMIC YEAR 2026 - 2027
Manage Student & Lecturer User Accounts
SECURITY POLICY
RBAC Strict Mode • bcrypt Hashing Enabled
Total Accounts: 1,248 Users
System Administrator
ID: AD202601
Manage User Accounts
Create student and lecturer accounts, edit user details, and lock or unlock user access immediately.
CREATE NEW USER ACCOUNT
ACCOUNT ROLE
Student
STUDENT ID / LECTURER ID
e.g., SV2026099 (type 'SV2026001' to test...
FULL NAME
Enter full name
INSTITUTIONAL EMAIL
user@university.edu
DEPARTMENT / MAJOR
Management Information Systems
INITIAL PASSWORD (BCRYPT)
Enter initial password
+ Create User Account
EXISTING USER ACCOUNTS DIRECTORY
Click Lock/Unlock to toggle account status immediately
Search by User ID, name, or email...
All Roles (Student & Lecturer)
All Account Statuses
USER ID
FULL NAME
ROLE
DEPARTMENT / MAJOR
EMAIL ADDRESS
PASSWORD
ACCOUNT STATUS
ACTIONS
SV2026001
Alex Nguyen
STUDENT
Management Information Systems
alex.nguyen@student.edu
A12331444141
ACTIVE
Edit
Lock Account
GV202601
Dr. Sarah Jenkins
LECTURER
Management Information Systems
sarah.jenkins@university.edu
macaro#12344
ACTIVE
Edit
Lock Account
GV202602
Prof. Michael Chen
LECTURER
Software Engineering
michael.chen@university.edu
Bilillyq21
ACTIVE
Edit
Lock Account
SV2026044
Tran Quoc Tuan
STUDENT
Computer Science
tuan.tran@student.edu
letgooooo123
LOCKED
Edit
Unlock Account

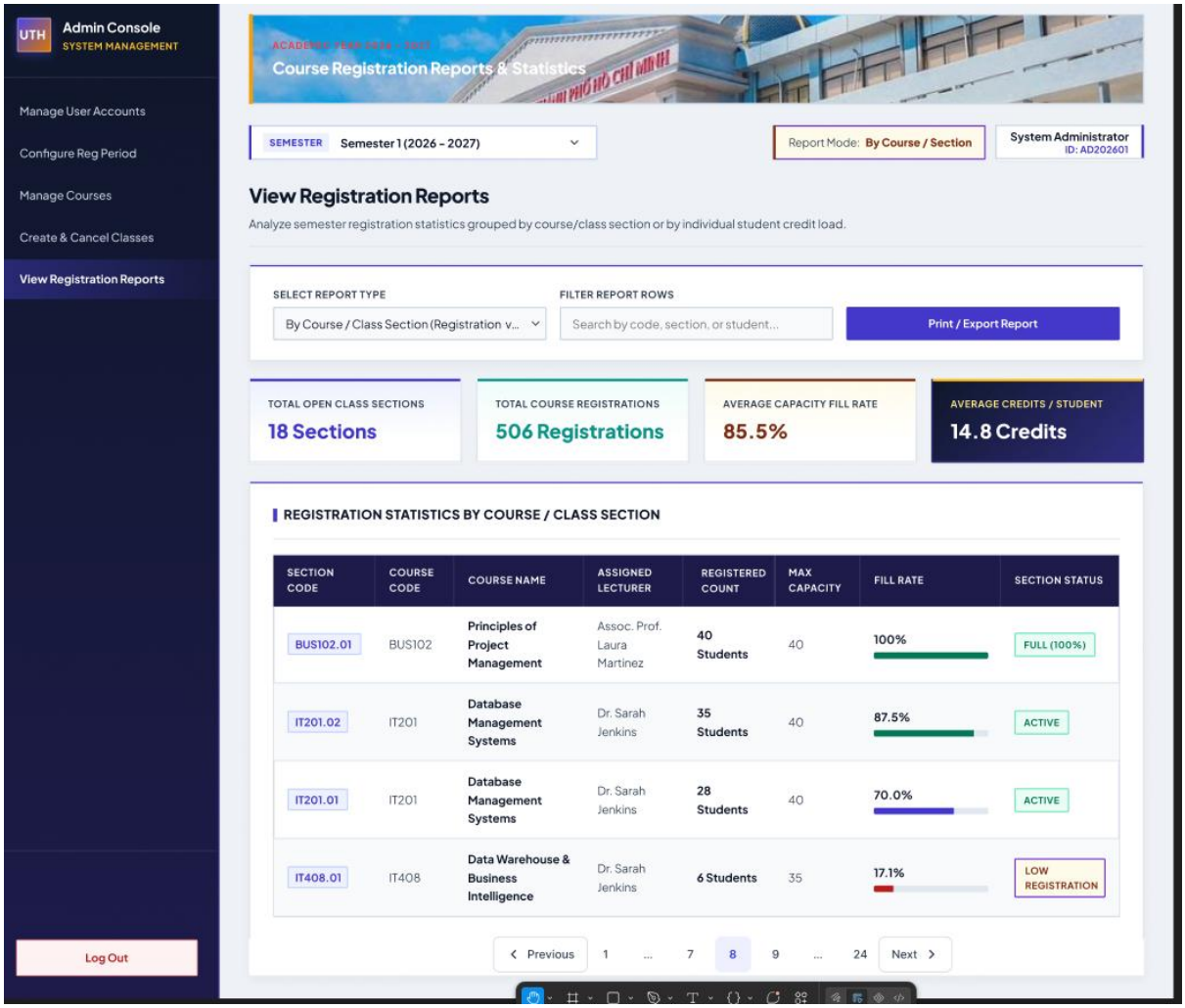
Manage Courses



Configure Registration Period



View Registration Reports



Manage Class Sections

